

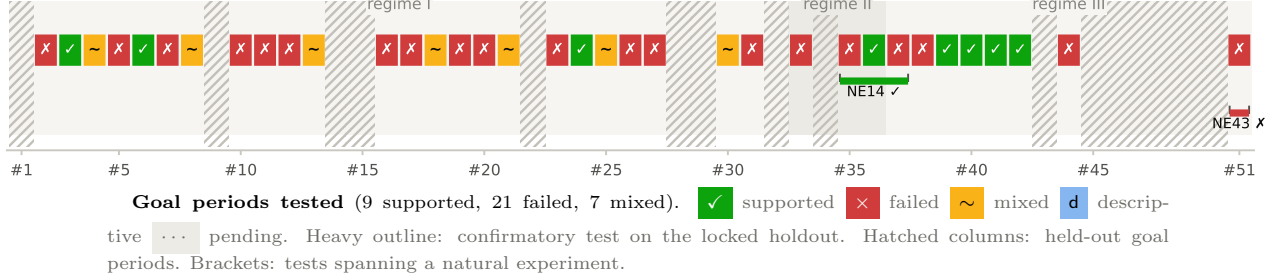
# H38: Joint silences are platform stalls

Vivian Rogers, with Claude Opus 5.5 · agent-swarm physics · Updated 2026-10-04

Complete 50% Faithfulness 2/5 Usefulness 2.5/5

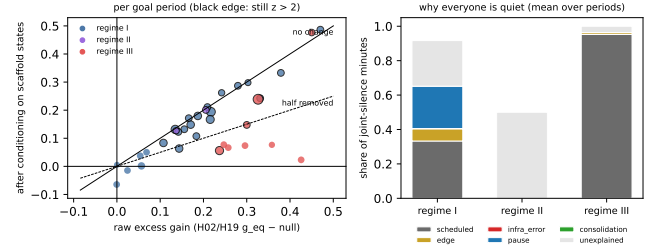
**Bottom line.** Joint silences are rare and operator-scheduled, not platform stalls; in the always-on regime 70–80% of apparent co-activation is the runner starting and stopping everyone together, so trim to the common running window before measuring synchrony.

level: hypothesis · pipeline checks 86% · updated 2026-10-04



## WHAT HAPPENED

Built a shared outage table (minutes with  $\leq 1$  active agent, causes from logs) and asked, in 35 non-holdout periods, how much of H02/H19's collective co-activation is infrastructure. **Round 1b** re-ran everything on the corrected activity table (round 1 had lost half of all events) with DQ8's calibrated null. **Joint silences are rare and scheduled:** 5.3% of minutes (was 14.3%), 78% outside the operator's run; error bursts make them *less* likely. **In always-on regime III most co-activation is day-edge synchrony** ( $f = 0.72$ ; 0.83 after trimming), and once each day is trimmed to the common running window only #44 and #51 (and #37, marginally) keep a significant excess; regime I keeps it in 16/18 periods. Native test NE43: the edge share does not drop when the operator's pause/resume messages stop, so the edges are the runner's schedule. Holdout not run.



**Observables.** Corrected tables. (a) Excess equal-time gain per goal period before and after conditioning on scaffold states (dashed: half removed). Regime-III points (red) drop to or below the dashed line except #44 and #51; regime I (blue) stays near the identity line. Black edge: still  $z > 2$ . (b) Causes of joint-silence minutes, mean over periods: with the dropped events restored, operator-scheduled minutes dominate; infrastructure errors are about 0.1%.

## PHYSICAL MODEL

Models 01 (Curie–Weiss) and 09 (Hawkes).  $s_i(t) = \pm 1$ : agent  $i$  active in minute  $t$ ;  $h_i(b)$ : per-agent field per 30-min block;  $z(t)$ : stall indicator (operator off, not yet started or finished, consolidating, infrastructure error), a field that switches agents off;  $\pi$ : stall fraction;  $p$ : on-rate. Stall-adjusted gain: trim to the window in which all agents run, set scaffold-unavailable agent-minutes to the block mean, compare with block-shift surrogates drawn on the kept rows.

$$P(\mathbf{s}) \propto \exp \left[ \frac{\beta J_0}{2N} M^2 + \sum_i (h_i(b) + \Delta_i z(t)) s_i \right], \quad M = \sum_i s_i$$

$$\text{VR} = \frac{\sum_b n_b \text{Var}_b M}{\sum_b n_b \sum_i \text{Var}_b s_i}, \quad g = 1 - \text{VR}^{-1}$$

stalls alone:  $\text{VR} - 1 \approx \frac{(N-1)\pi p}{1 - (1-\pi)p}$  ( $N=12, p=0.5, \pi=0.05 \Rightarrow g \approx 0.34$ )

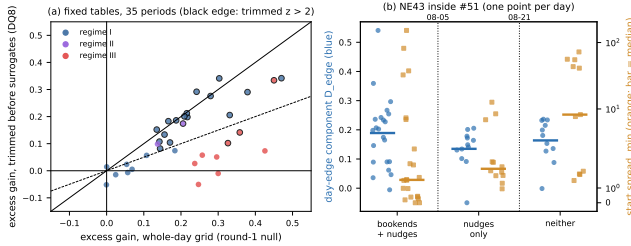
$$f = 1 - \frac{g_{\text{adj}} - \langle g_{\text{adj}} \rangle_{\text{null}}}{g_{\text{raw}} - \langle g_{\text{raw}} \rangle_{\text{null}}}$$

## SO WHAT? PRACTICAL VALUE FOR UNDERSTANDING AGENT SWARMS

A useful negative plus one practical rule, now sharper. If you run an always-on swarm with a daily start and stop, any synchrony number (mean-field coupling, market-mode eigenvalue, co-activation) is mostly your own runner schedule: agents boot and stop together whether or not you announce it. Trim each day to the window in which all agents run *before* drawing surrogates; here that removes about 80% of regime-III co-activation and leaves real excess only where agents genuinely co-move (#44, #51). Joint silences are not a stall signal worth alerting on, error logs don't predict them, and provider outages leave no trace. Message-driven swarms (regime I) get nothing from this.

## RESULTS VS PREDICTIONS

| Prediction                                                                  | Round 1 → round 1b (corrected tables)                                               | Verdict |
|-----------------------------------------------------------------------------|-------------------------------------------------------------------------------------|---------|
| P1 JS share 0.05–0.30, median 0.10 → 0.009; excess 0.008 above independence | → 0.007                                                                             | ×       |
| P2a scaffold reasons in JS above chance                                     | explained 0.51 → 0.88 (mostly scheduled); > surrogate in 23 → 24/35                 | ×       |
| P3 infra bursts precede JS                                                  | log OR $-0.43 \rightarrow -0.96$ ; significant > 0 in 0 → 1/30                      | ×       |
| P4 stall filter removes $\geq$ half                                         | $f$ 0.25 → 0.26 (III 0.63 → 0.56)                                                   | ×       |
| Amend. 2 scaffold conditioning                                              | $f_{III}$ 0.68 → 0.72; $f_I$ 0.11 → 0.11                                            | III ✓   |
| DQ8 null (trim first)                                                       | significant 29 raw → 20/35 new trimmed; III 3/8, I 16/18                            |         |
| P5 $\lambda_1$ drop tracks stall share                                      | $\rho$ 0.89 → 0.87; ratio 0.48 → 1.00; $\sim \rightarrow$ trimmed edge: 19/41 units | ✓       |
| P6 III–I contrast shrinks $\geq$ half                                       | +0.125 → +0.017; now +0.155 → −0.006                                                | ✓       |
| P9 NE14 jump carried by stalls                                              | +0.15 → −0.06; now +0.12 [−0.04, 0.27] → −0.14                                      | ✓       |
| NE43 bookends stop (native)                                                 | edge share 0.19 → 0.14 ( $p$ 0.08); start spread narrower                           | ×       |



**Second figure.** (a) Excess gain on the whole-day grid (round-1 null) vs after trimming to the all-present window before drawing surrogates (DQ8): regime III collapses toward zero, regime I stays. (b) NE43 inside #51, one point per day: the day-edge component (blue) does not fall when the bookend messages stop on 08-05; the start spread (orange) widens only when the nudger stops on 08-21 (post hoc).

## GOAL PERIODS IN DETAIL

G02 × failed  
 G03 ✓ supported  
 G04 ~ mixed  
 G05 × failed  
 G06 ✓ supported  
 G07 × failed  
 G08 ~ mixed  
 G10 × failed  
 G11 × failed  
 G12 × failed  
 G13 ~ mixed  
 G16 × failed  
 G17 × failed  
 G18 ~ mixed  
 G19 × failed  
 G20 × failed

+21 more periods: see the card.

## FAITHFULNESS SCORECARD

|                        |   |                   |   |
|------------------------|---|-------------------|---|
| A mapping              | 1 | F identifiability | 1 |
| B assumptions          | 1 | G ground truth    | 1 |
| C adequacy             | 1 | H comparative     | 1 |
| D unfitted predictions | 1 | I transfer        | 1 |
| E interventional       | 1 |                   |   |

Level: **hypothesis**. Scale 0 not done or failed, 1 partial, 2 passed (`writeup/paper.tex`).

## CAVEATS

LLM API failures leave no log line, so provider outages remain untested. On the corrected tables most periods have only a handful of joint-silence minutes, so the per-period “explained above chance” test (and 12 per-period verdict changes, mostly regime I) rests on few minutes. The headline estimator (scaffold conditioning) was added after synthetic smoke runs; the pre-registered stall filter failed. The **scheduled** flag is built from operator messages and empties after 08-05; agent-level trimming does not. A whole-block stall (the 300-min stop on 2025-06-18) leaves no trace in the block-detrended gain. The holdout has not been run.

## ROUND 2

Per-provider silence test (all agents of one lab silent while others act) for API outages; run `confirm.py` with the trimmed (DQ8) variants added before it is executed; study the #44 and #51 residual coupling; replace the message-based **scheduled** flag by the runner’s start/stop times.

*No round-2 redirects yet.*